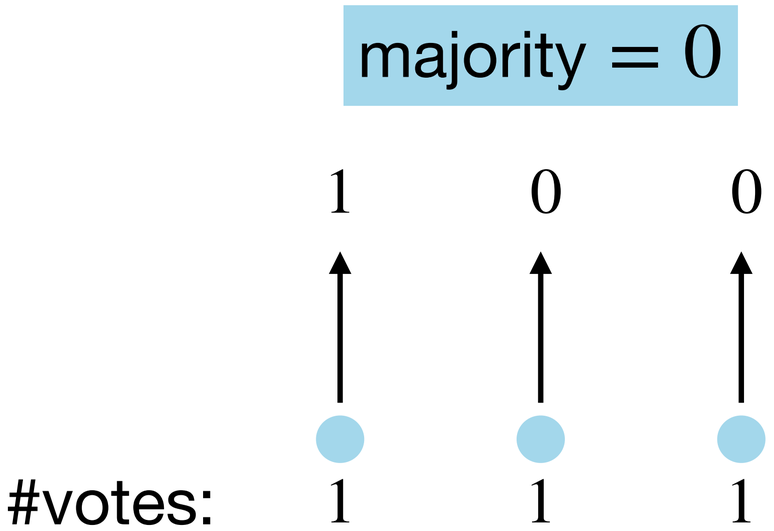
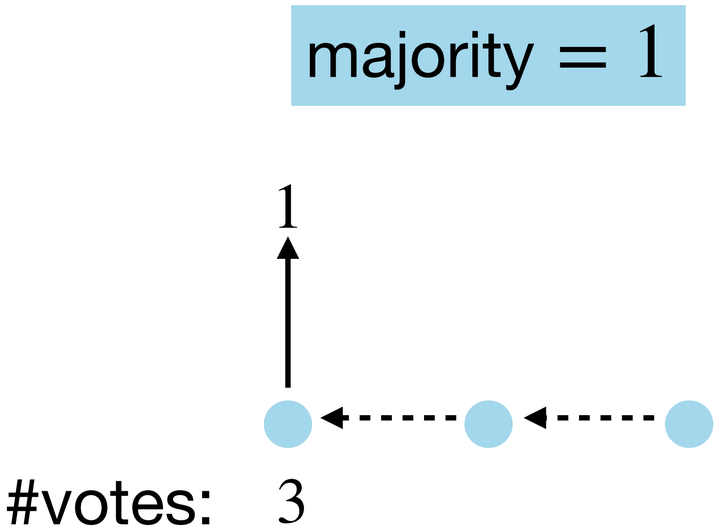
Fairness in liquid democracy

Michelle X. Yeo

Direct vs liquid democracy





Why liquid democracy?

- Practicality/efficiency:
  - Increasingly used in governance, corporations, asset management, and blockchain settings
  - Faster decision making in DAOs in blockchains

German Pirate Party [KKHSS21]

Delegated Proof of Stake

Why liquid democracy?

- Fairness:
  - (Individual): most ‘ignorant’ voters abstain, a way to encourage them to participate
  - (Societal): better decisions when voters delegate to more informed voters

Better decisions?

- What does better mean?
- How is a decision made?
- Is this always true? (When is this true?)

Better decisions?

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Spoiler: no!

Talk Outline

1. Liquid democracy with ground truth: on the manipulation of variance

2. Liquid democracy with opposing factions

Talk Outline

1. Liquid democracy with ground truth: on the manipulation of variance

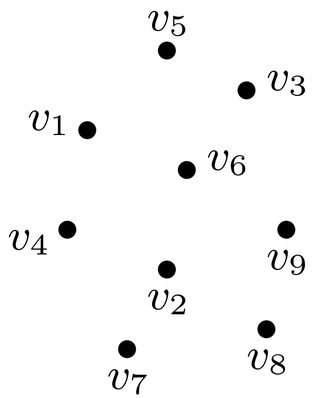
2. Liquid democracy with opposing factions

Model and setting

- Binary voting with ground truth
- $G = (V, E, \mathbf{p})$

Competencies:

$p_1 = 0.8$
$p_2 = 0.6$
$p_3 = 0.5$
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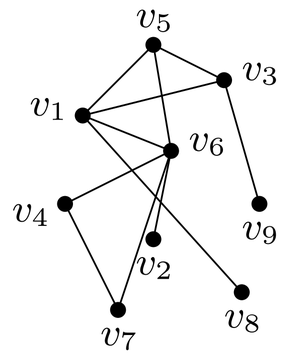


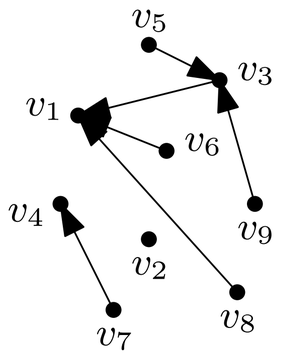
Model and setting

- Binary voting with ground truth
- $G = (V, E, \mathbf{p})$
- Local delegation mechanisms
- $\alpha :=$ approval threshold
- $J(i) :=$ approval set of v_i

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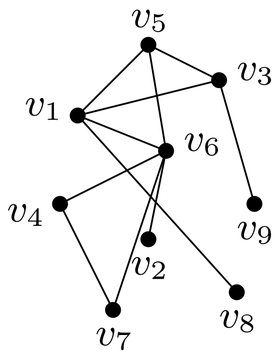
$x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9$

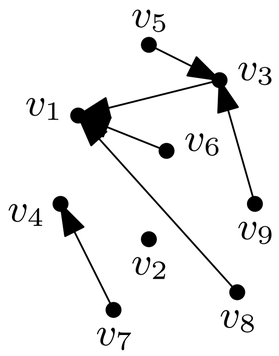
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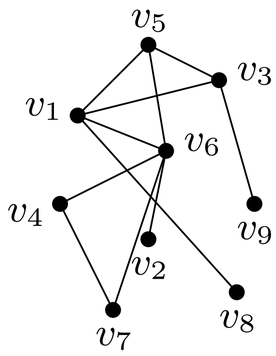
Main mechanism:

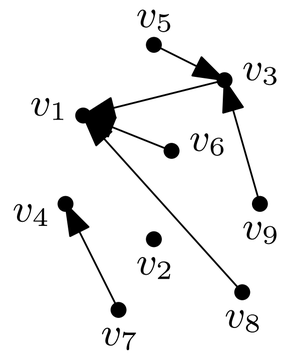
If $|J(i)| > t$ then v_i delegates to random voter in $J(i)$

Probability of correct decision

- Delegation graph: output of applying mechanism to input graph
- w_i weight of ith sink counts number of votes delegated
- Correct decision chosen if
$$\sum_{v_i \in S'} w_i \geq \sum_{v_i \in S \setminus S'} w_i$$
- $P^M(G)$: Probability of correct decision

Competencies:
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$x_1, x_2, x_3, x_4, x_5, x_6, x_7, x_8, x_9$

Desiderata

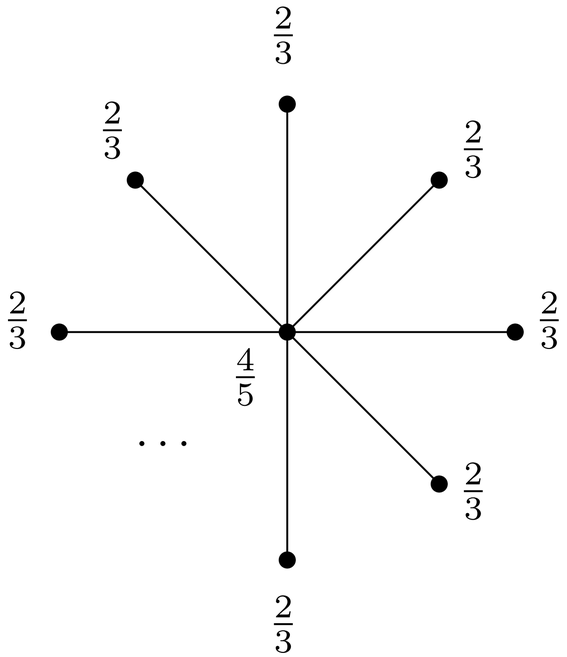
$G = (V, E, \mathbf{p})$

- Positive gain: **exists** some instances where delegation performs better
- Do no harm: **for all** instances, the loss goes asymptotically goes to 0 (as the graph grows larger)
- **Strong positive gain**: **for all** instances, delegation performs better

Gain: $P^M(G) - P^D(G)$

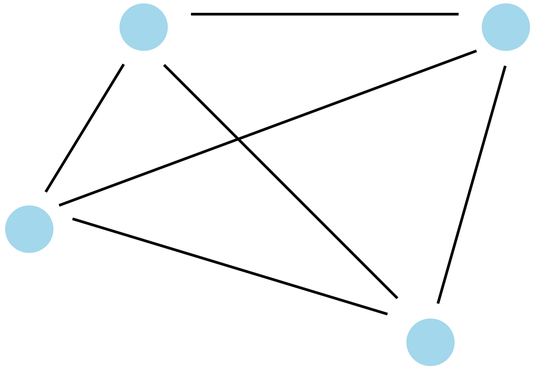


Better decisions?



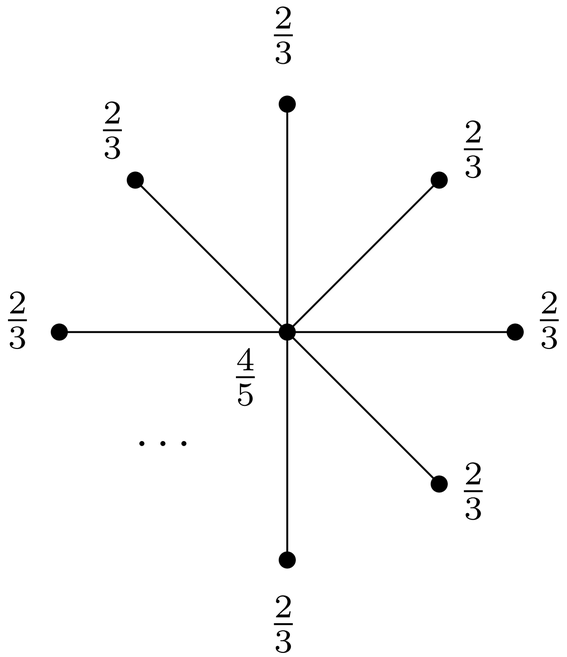
[KMP21]

Complete graph



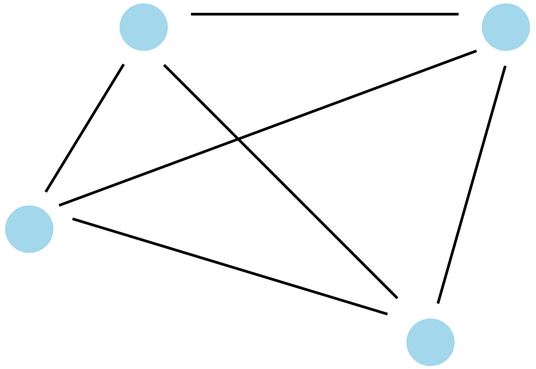
[HHJMPR23]

Better decisions?



[KMP21]

Complete graph

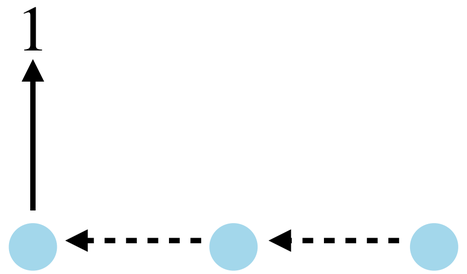


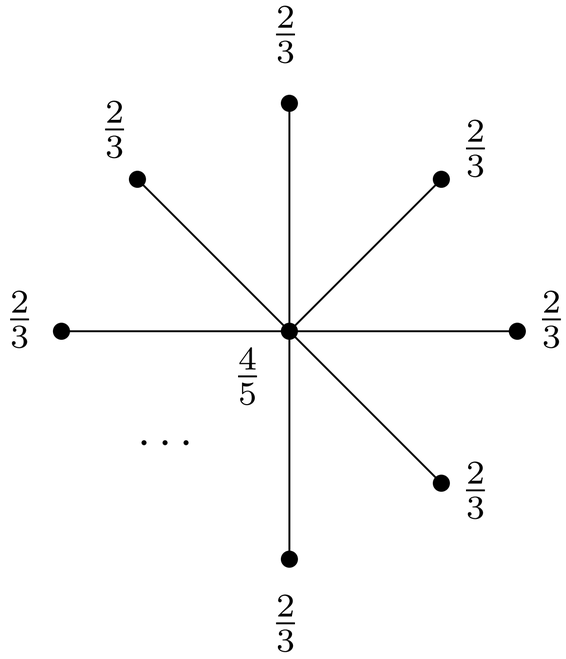
[HHJMPR23]

Our work: positive results for complete, random d-regular, bounded degree, and bounded minimal degree graphs

Why our setting is hard

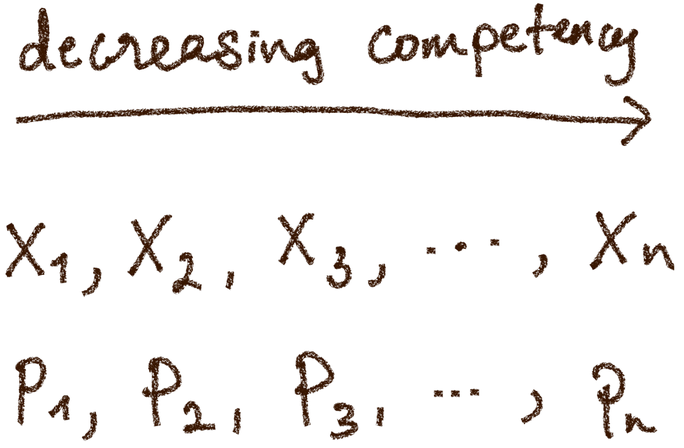
1. Positively correlated random variables
    - Delegation correlates voting outcome
    - Exist extensions of Chernoff bound for negatively correlated random variables but not positive
2. Need sufficient variance among voting outcomes to achieve Do no harm
    - Unusual in algorithm design!





Recycle sampling

- Bound the sum of **positively correlated** random variables
- Predecessor variables have higher competency



v_6 delegates to $v_1 \Rightarrow P_6$ replaced by P_1

Recycle sampling

- Bernoulli parameter can be recycled from predecessors
- Partition: set of voters where there are no edges between voters
- First partition: all vote independently
- Partition complexity c: number of partitions
  - Depends on approval threshold α

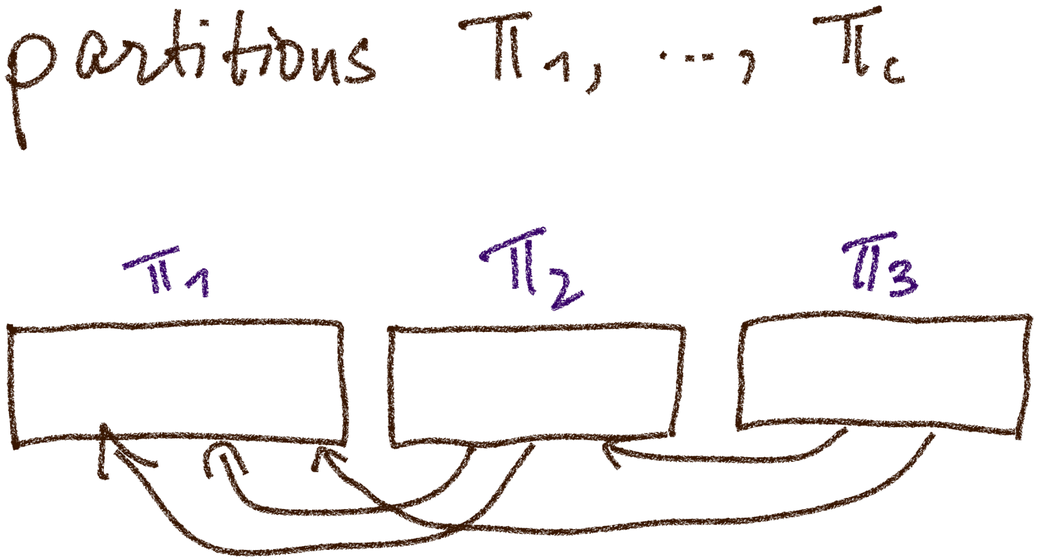
partitions $\pi_1, \dots, \pi_c$



↳ longest delegation path

Recycle sampling

- x_i random variable that represents delegation
  - w.p. z_i it is a Bernoulli random variable with parameter p_i
  - w.p. $(1 - z_i)$ takes the outcome of random successor
- Recycle sampled random variable $X_n = \sum_{i=1}^{n} x_i$ associated with G



Recycle sampling

Theorem

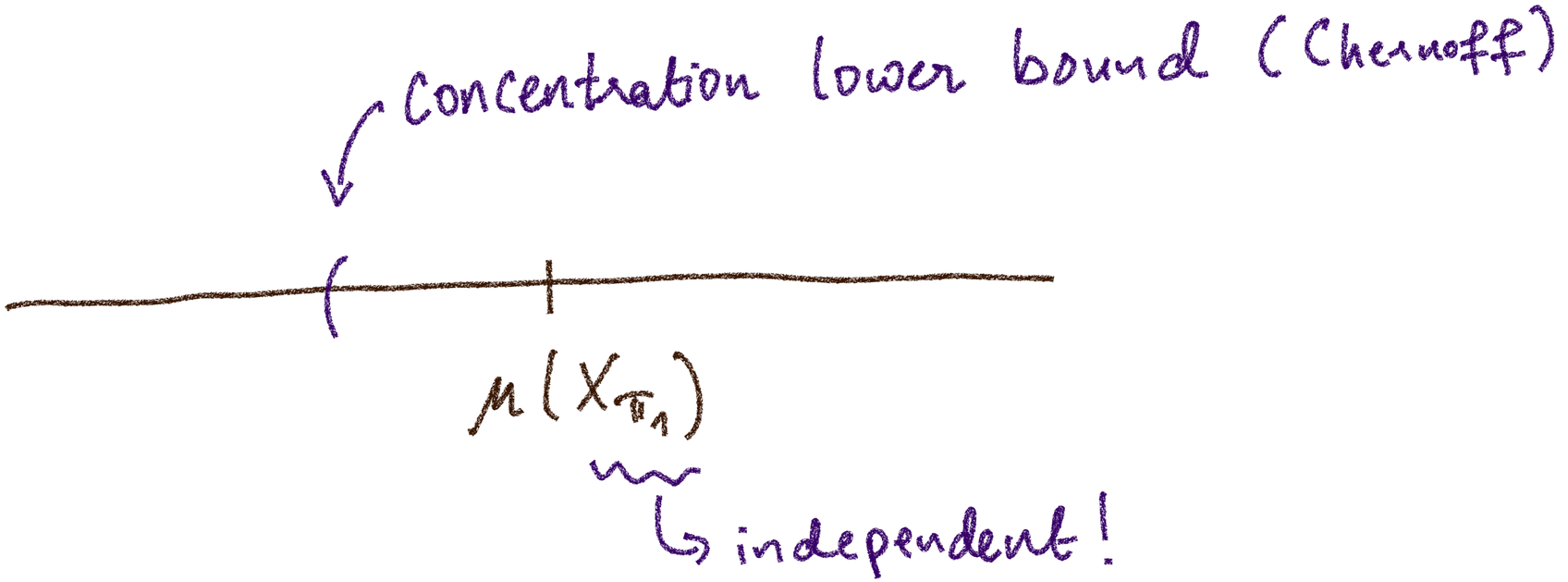
Let X_n be a random variable associated with a recycle sampled graph G. Then, with probability at least $1 - e^{-\Omega(j^{1/3})}$,

$$X_n \geq \mu(X_n) - c \cdot \frac{\varepsilon n}{j^{1/3}}$$

↳ degree of dependency.

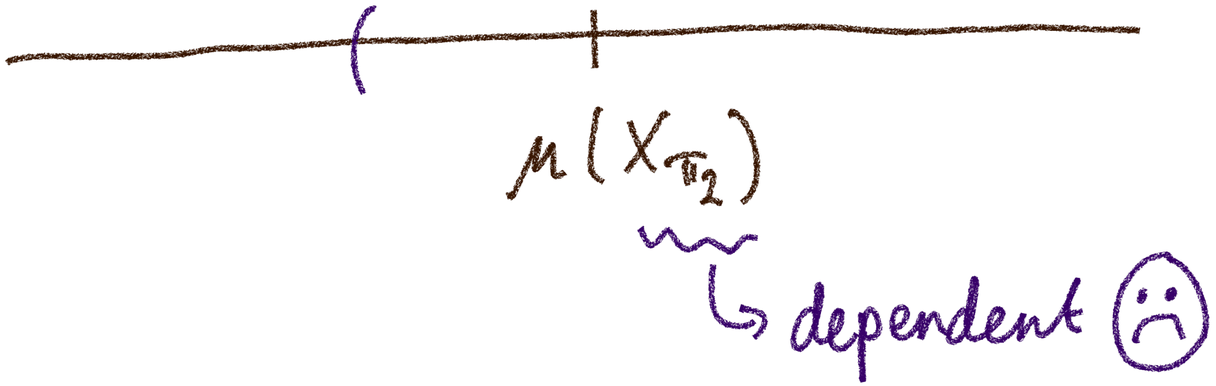
Recycle sampling

- Chernoff-like bound for sum of recycle sampled random variables from the mean



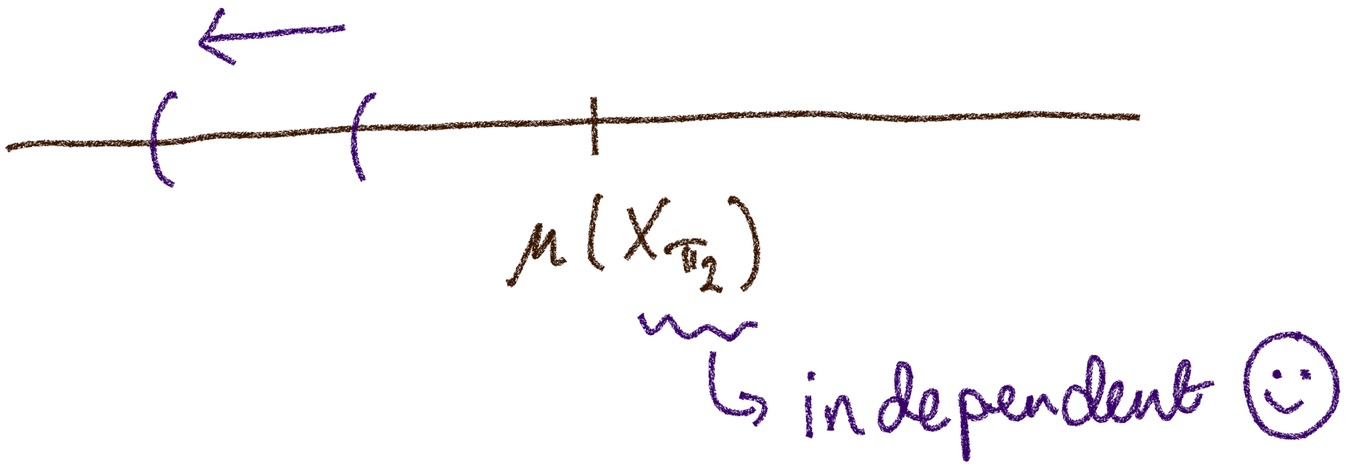
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Recycle sampling

- Chernoff-like bound for sum of recycle sampled random variables from the mean



Transmutation[3]: manipulating variance

- Intuition: sufficient variance in voting outcome helps to ensure DNH!
- Two conditions to transmute variance:
  1. $\mathbf{p} \in (\beta, 1-\beta)$
  2. maximum weight of sink



[3]DnD 5e handbook: “You are a student of spells that modify energy and matter”

Results

Theorem (informal)

There are local delegation mechanisms such that strong positive gain and do no harm are satisfied on complete and random d -regular graphs

Results

Theorem (informal)

There are local delegation mechanisms such that strong positive gain and do no harm are satisfied on bounded degree and bounded minimal degree graphs

Fairness implications

- Concentration of influence can lead to harm (anti-dictator)
- Graphs with not too much structural asymmetry (degrees)
- Best case: complete graph (everyone knows and trusts everyone)

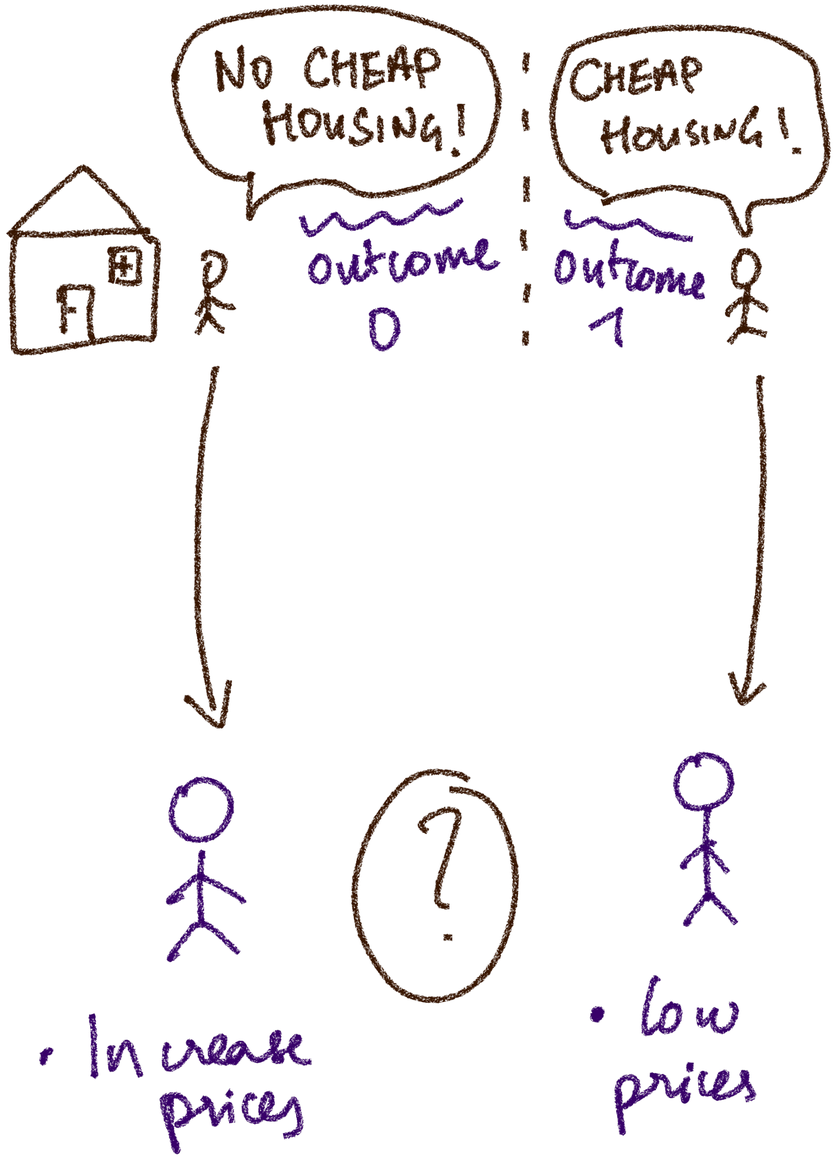
Talk Outline

1. Liquid democracy with ground truth: on the manipulation of variance

2. Liquid democracy with opposing factions

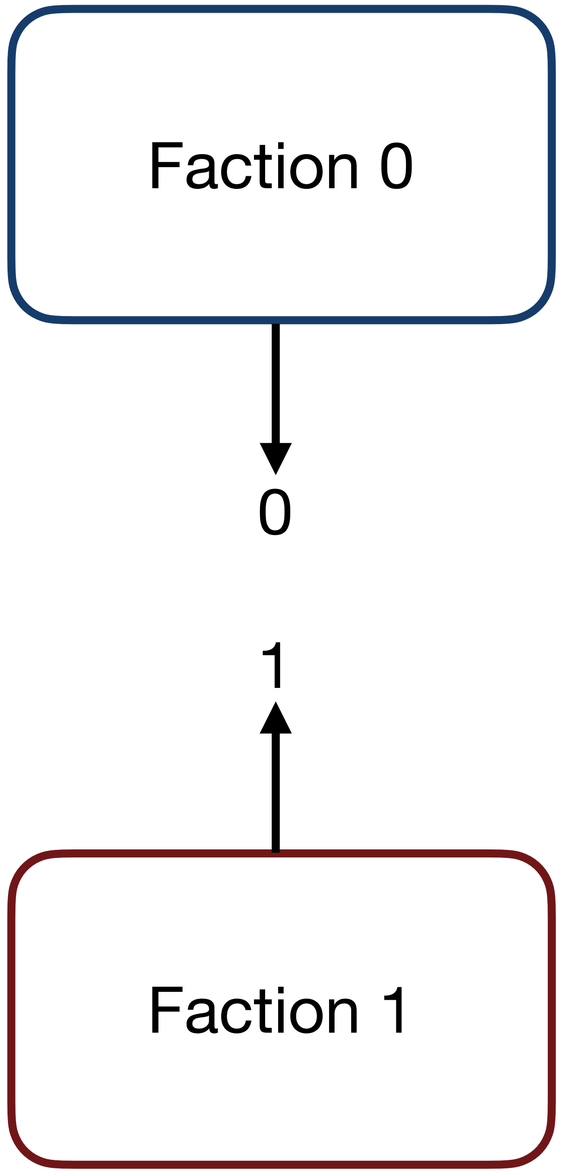
No more ground truth!

- Voters have preferences $\in \{0,1\}$ for outcome
- Competencies: how likely voters will vote for preference
  - Voter indecision
  - General stochasticity between casting a vote and desired outcome



No more ground truth!

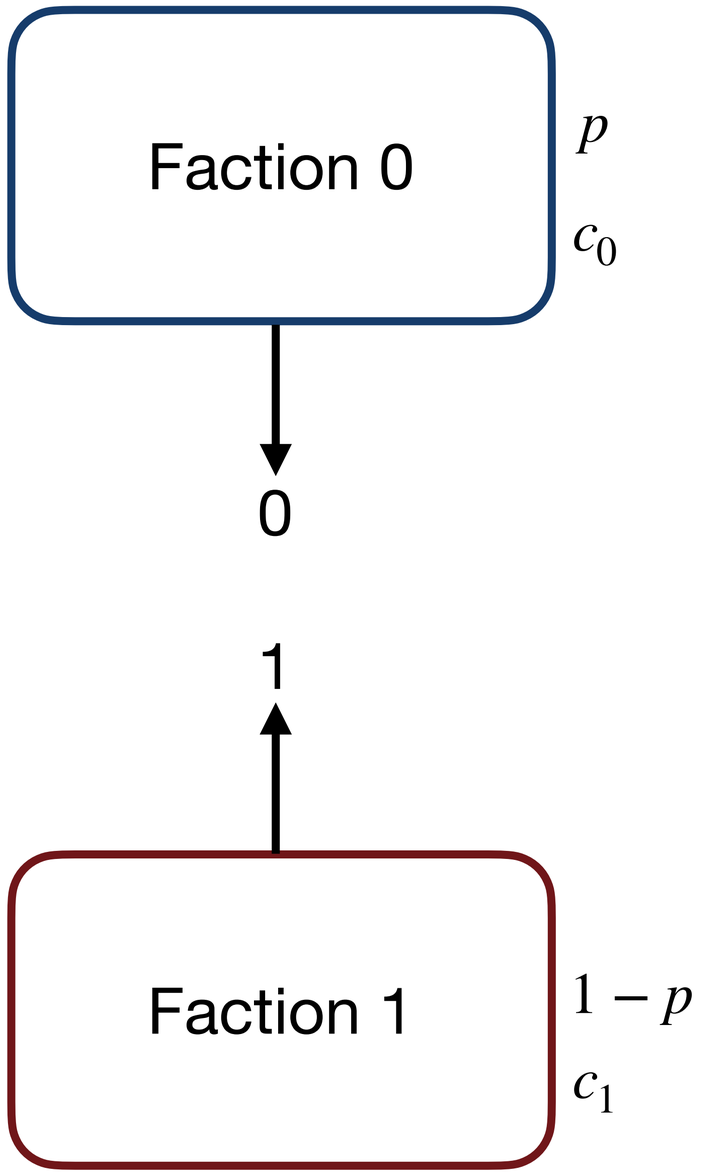
- Voters have preferences $\in \{0,1\}$ for outcome
- Competencies: how likely voters will vote for preference
  - Voter indecision
  - General stochasticity between casting a vote and desired outcome
- Faction goal: design delegation mechanisms to maximise probability of winning election



Does delegation even help?

Theorem

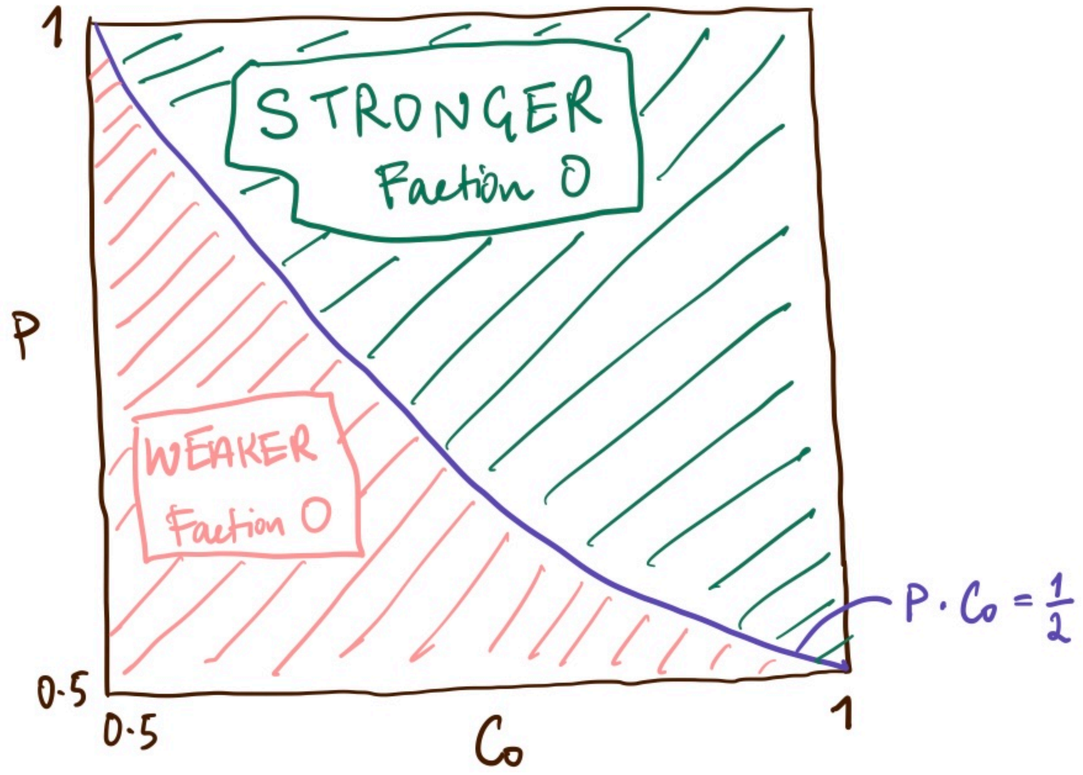
Finding optimal strategies in general setting is PSPACE-complete



- Simplified setting with parameters p, c_0, c_1
- $c_0, c_1 > 0.5$
- Wlog focus on strategies for faction 0
- Possible strategies: Dictator, Direct, Representative(k)

Strength of faction 0

- Assume $c_1 = 1$ (strongest possible opponent)



Equal proportion

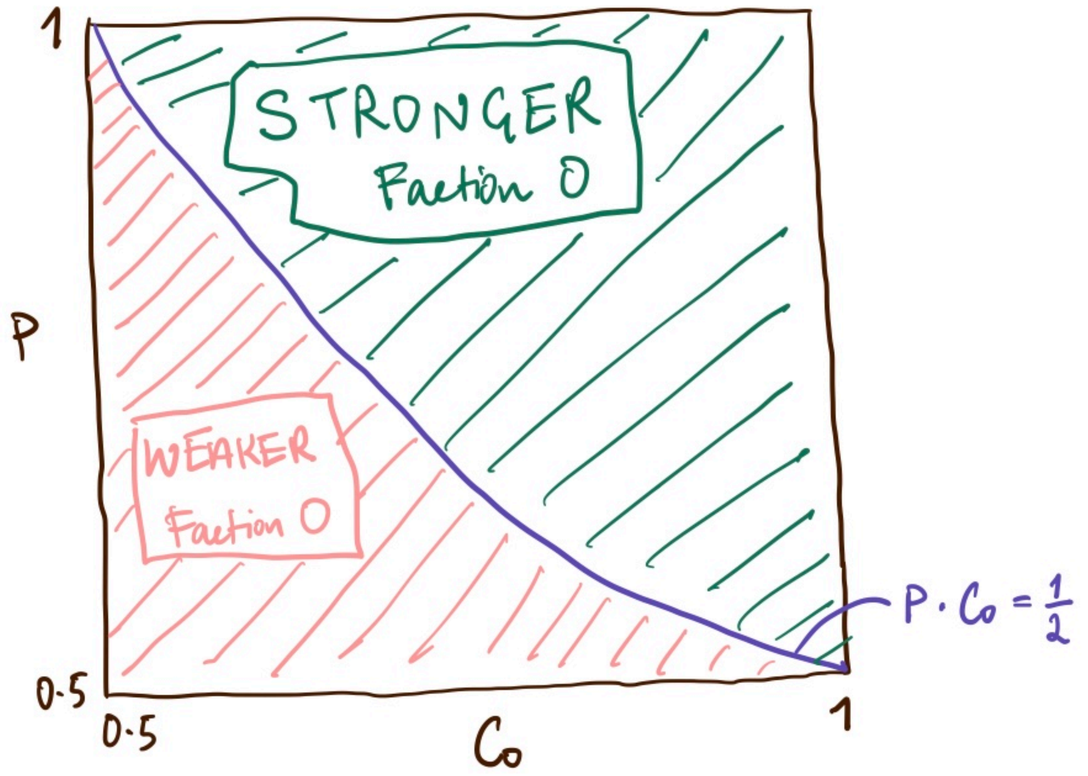
- $c_1 = 1, p = 0.5$

Theorem (informal)

Both factions playing Dictator is the unique Nash Equilibrium.

Faction 0 is stronger

- $c_1 = 1,\ p \cdot c_0 \geq 0.5$

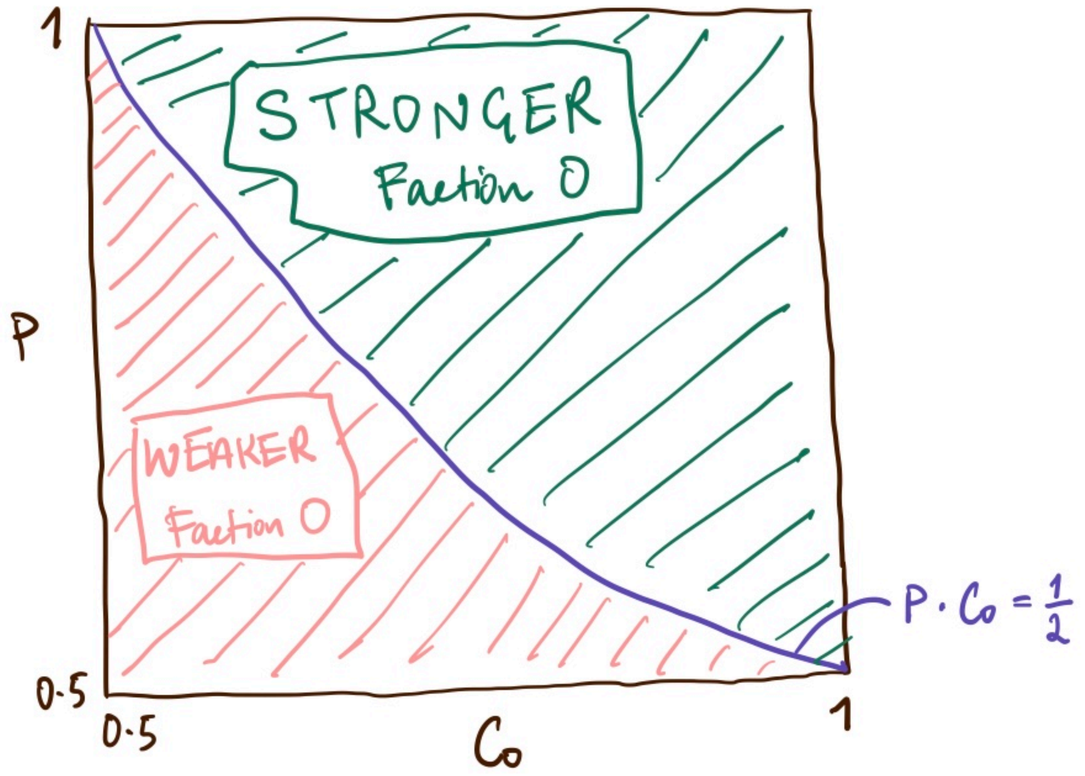


Theorem (informal)

Faction 0 wins over faction 1 by playing Direct.

Faction 0 is weaker

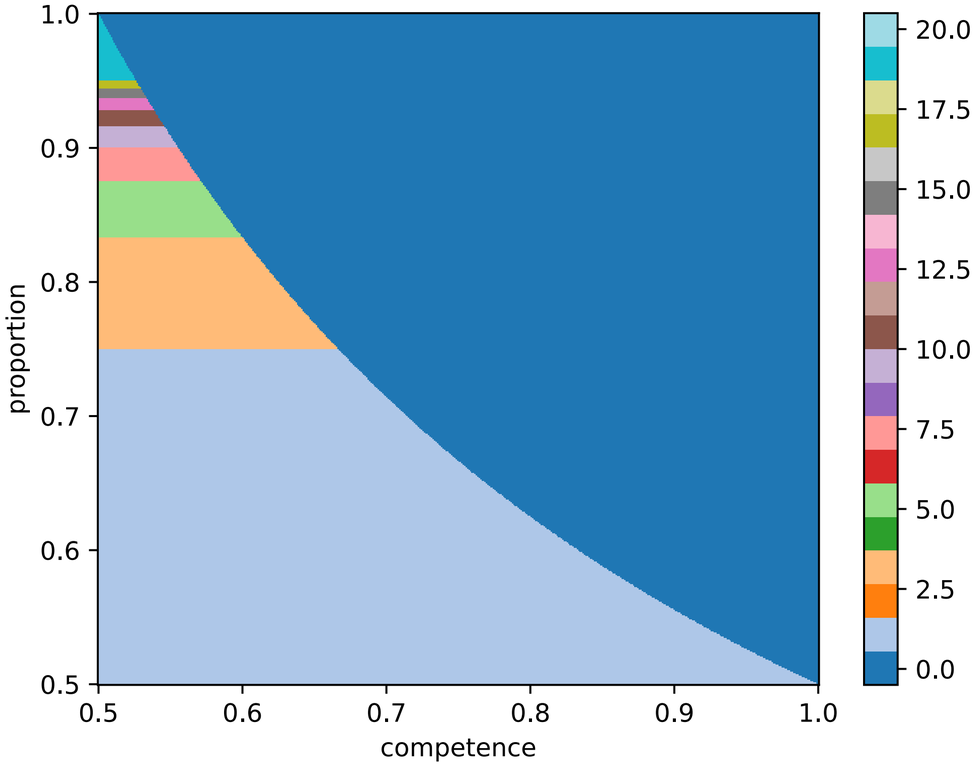
- $c_1 = 1,\ p \cdot c_0 < 0.5$



Theorem (informal)

Delegation improves winning probability over direct.

Proportion vs. competency



Fairness implications

- Election fairness: weaker party should win once in a while (proportionate to power in blockchain settings)
- Direct voting is unfair sometimes!
- Act of delegation directly improves fairness for weaker party (vs direct voting!)
- Delegation compensates for lower competency by size

Conclusion

- 2 perspectives on delegation: with ground truth and without
  - With ground truth: variance and connectivity matters!
  - Without ground truth: delegation achieves fairness for weaker party (large in size but less competent)

michellexyeo@gmail.com

Ongoing and future work

- Can delegation improve security against misinformation/fake news?
- Complex delegation functions
- Private delegations using MPC
- Consent and delegation (“The Killer Vote”)
- Accountability in delegation